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# ***Problem Solving with AI***

## ***(Claude Code + JavaScript)***

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### **Q1. JavaScript Coding Practice**

#### **1. Convert Celsius to Fahrenheit**

Write a function that takes a temperature in Celsius and converts it to Fahrenheit using the formula:  $F = (C \times 9/5) + 32$ .

#### **Source code:**

```
function celsiusToFahrenheit(celsius) {  
    let fahrenheit = (celsius * 9 / 5) + 32;  
    return fahrenheit;  
}  
  
let celsius = 25;  
let result = celsiusToFahrenheit(celsius);  
console.log("Celsius:", celsius);  
console.log("Fahrenheit:", result);
```

#### **Input;**

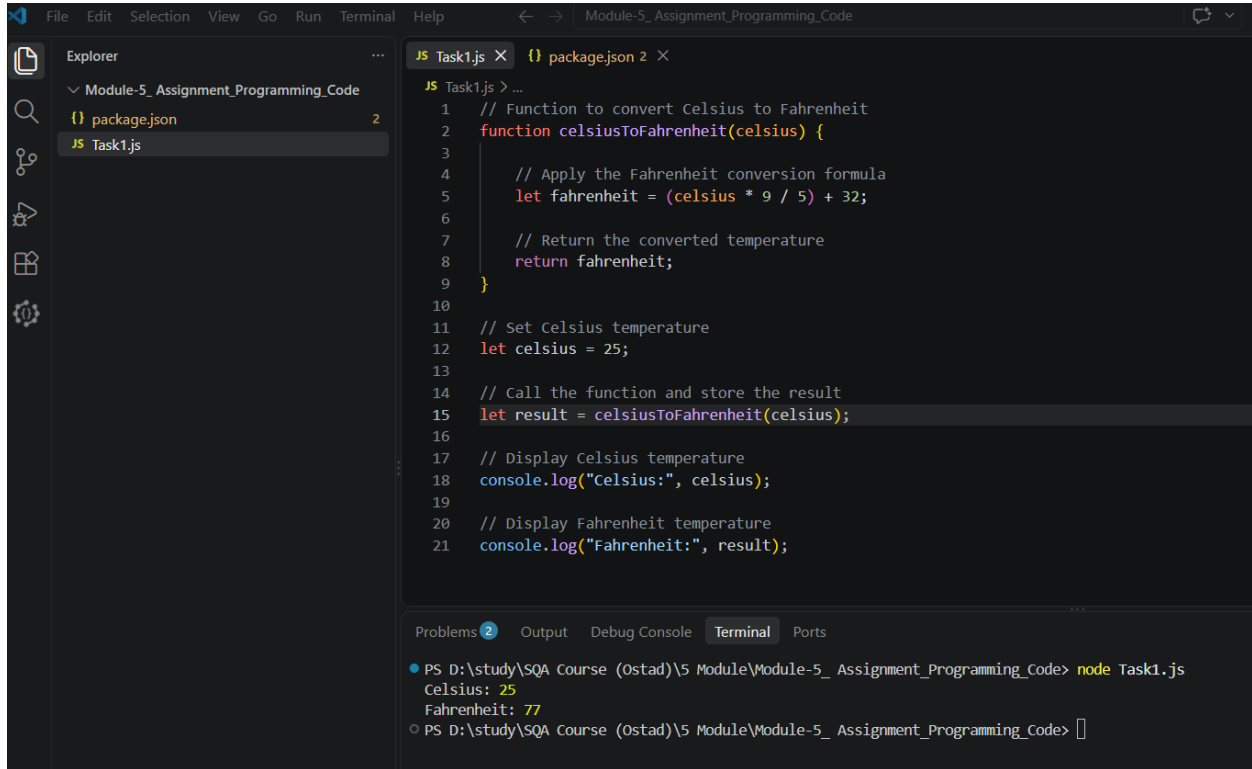
Celsius = 25

#### **Output:**

Celsius: 25

Fahrenheit: 77

## Screenshot:



The screenshot shows the Visual Studio Code editor with a project named 'Module-5\_Assignment\_Programming\_Code'. The Explorer sidebar on the left shows the file structure with 'package.json' and 'Task1.js'. The main editor window displays the code in 'Task1.js'.

```
1 // Function to convert Celsius to Fahrenheit
2 function celsiusToFahrenheit(celsius) {
3
4     // Apply the Fahrenheit conversion formula
5     let fahrenheit = (celsius * 9 / 5) + 32;
6
7     // Return the converted temperature
8     return fahrenheit;
9 }
10
11 // Set Celsius temperature
12 let celsius = 25;
13
14 // Call the function and store the result
15 let result = celsiusToFahrenheit(celsius);
16
17 // Display Celsius temperature
18 console.log("Celsius:", celsius);
19
20 // Display Fahrenheit temperature
21 console.log("Fahrenheit:", result);
```

The bottom panel shows the 'Terminal' tab with the following output:

```
PS D:\study\SQA Course (Ostad)\5 Module\Module-5_Assignment_Programming_Code> node Task1.js
Celsius: 25
Fahrenheit: 77
PS D:\study\SQA Course (Ostad)\5 Module\Module-5_Assignment_Programming_Code>
```

## Explanation:

- The function takes a temperature value in Celsius as input.
- It uses the formula  $F = (C \times 9/5) + 32$  to convert Celsius to Fahrenheit.
- The calculated value is stored in the fahrenheit variable.
- The function returns the converted Fahrenheit value.
- For example,  $25^{\circ}\text{C} = 77^{\circ}\text{F}$ .

## 2. Find the Factorial of a Number

Write a function that calculates the factorial of a given non-negative integer (e.g.,  $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$ ).

### Source Code:

```
function factorial(number) {  
    let result = 1;  
    for (let i = 1; i <= number; i++) {  
        result = result * i;  
    }  
    return result;  
}  
  
let number = 5;  
console.log("Number:", number);  
console.log("Factorial:", factorial(number));
```

### Input;

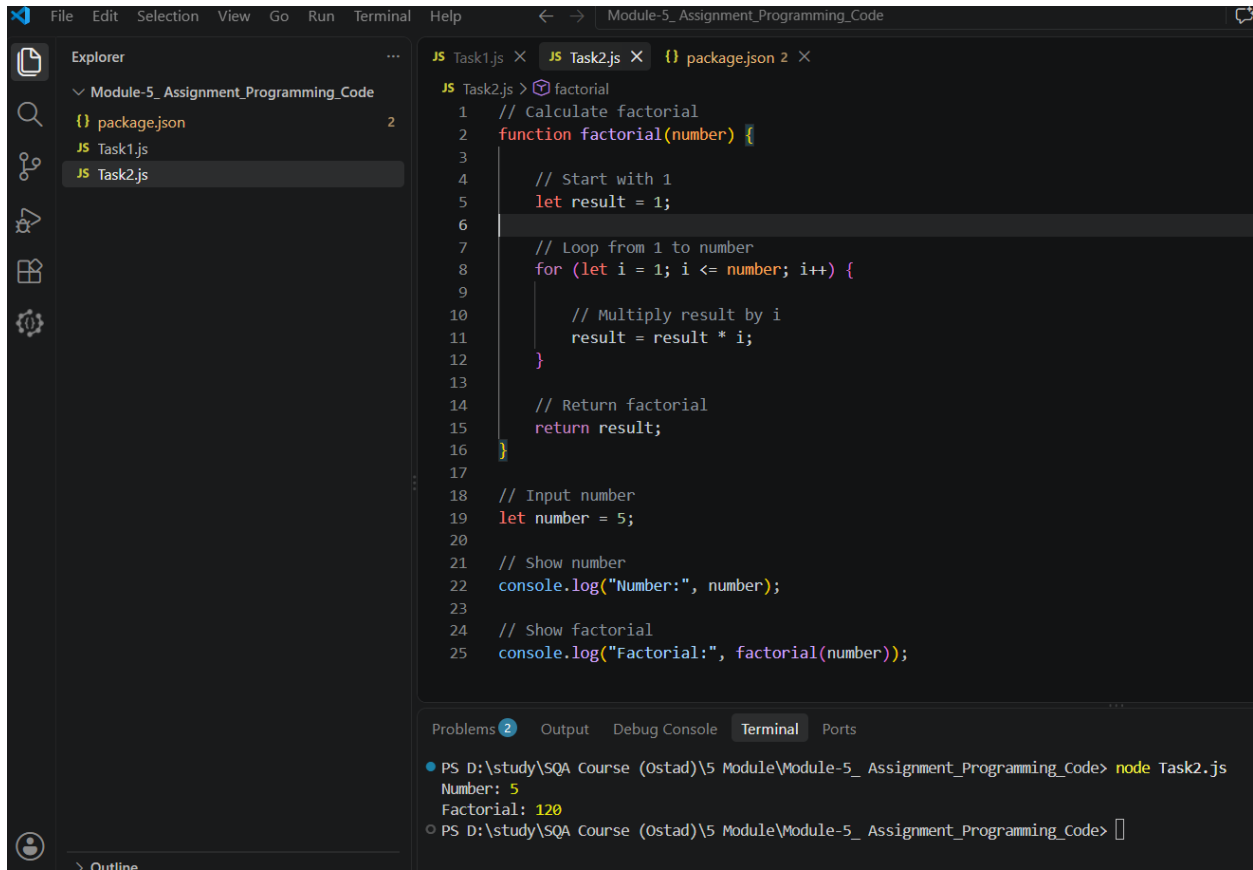
Number = 5

### Output;

Number: 5

Factorial: 120

## Screenshot:



The screenshot shows a Visual Studio Code editor window with a project named 'Module-5\_Assignment\_Programming\_Code'. The Explorer sidebar on the left shows the file structure with 'package.json', 'Task1.js', and 'Task2.js'. The main editor area displays 'Task2.js' with a JavaScript function to calculate the factorial of a number. The code is as follows:

```
1 // Calculate factorial
2 function factorial(number) {
3
4     // Start with 1
5     let result = 1;
6
7     // Loop from 1 to number
8     for (let i = 1; i <= number; i++) {
9
10        // Multiply result by i
11        result = result * i;
12    }
13
14    // Return factorial
15    return result;
16 }
17
18 // Input number
19 let number = 5;
20
21 // Show number
22 console.log("Number:", number);
23
24 // Show factorial
25 console.log("Factorial:", factorial(number));
```

At the bottom, the Terminal panel shows the output of running 'node Task2.js' in the directory 'PS D:\study\SQL Course (Ostad)\5 Module\Module-5\_Assignment\_Programming\_Code':

```
Number: 5
Factorial: 120
```

## Explanation:

- The function takes a non-negative integer as input.
- It starts with the result value as **1**.
- A for loop is used to multiply numbers from 1 to the given number.
- The final multiplied value is returned as the factorial.
- For example, **5! = 1 × 2 × 3 × 4 × 5 = 120**.

### 3. Check for Palindrome

Write a function that checks if a given word (e.g., "racecar") reads the same backward as forward and returns true or false.

#### Source Code:

```
function isPalindrome(word) {  
    let reversed = word.split("").reverse().join("");  
    return word === reversed;  
}  
  
let word = "racecar";  
console.log("Word:", word);  
console.log("Palindrome:", isPalindrome(word));
```

#### Input:

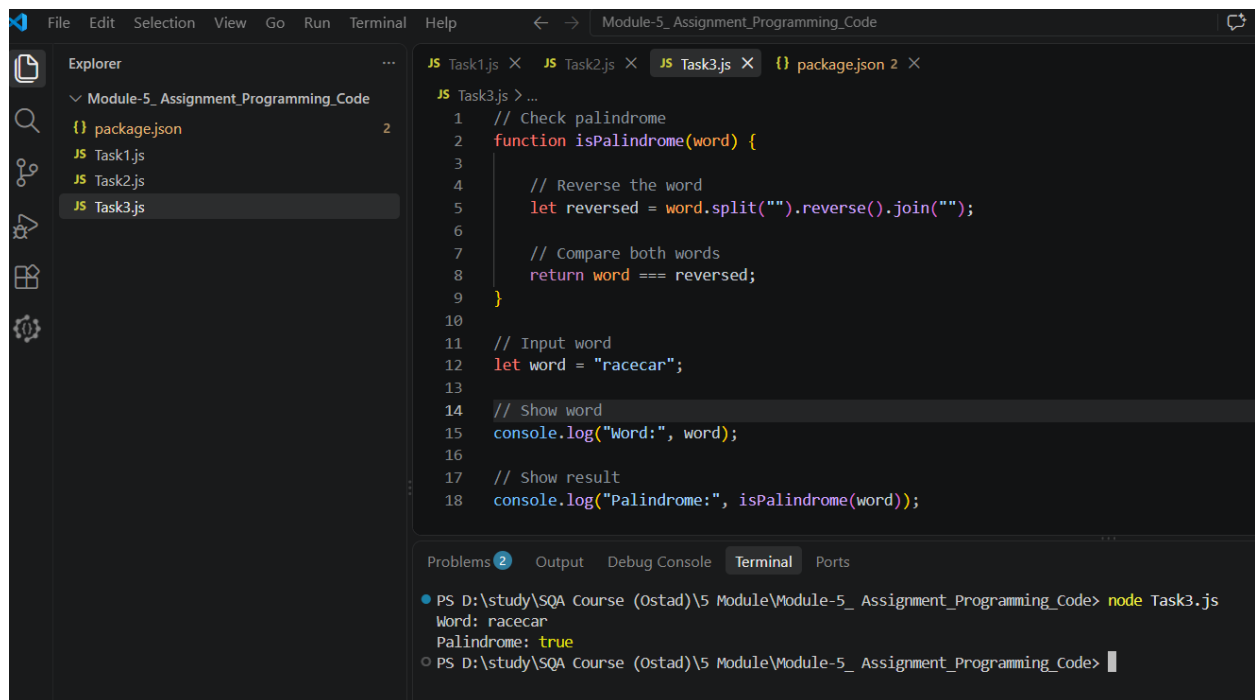
Word = racecar

#### Output:

Word: racecar

Palindrome: true

## Screenshot:



The screenshot shows the Visual Studio Code editor interface. The Explorer panel on the left displays the file structure for 'Module-5\_Assignment\_Programming\_Code', including 'package.json', 'Task1.js', 'Task2.js', and 'Task3.js'. The main editor area shows the code for 'Task3.js':

```
1 // Check palindrome
2 function isPalindrome(word) {
3
4     // Reverse the word
5     let reversed = word.split('').reverse().join('');
6
7     // Compare both words
8     return word === reversed;
9 }
10
11 // Input word
12 let word = "racecar";
13
14 // Show word
15 console.log("Word:", word);
16
17 // Show result
18 console.log("Palindrome:", isPalindrome(word));
```

The bottom panel shows the Terminal output:

```
PS D:\study\SQA Course (Ostad)\5 Module\Module-5_Assignment_Programming_Code> node Task3.js
Word: racecar
Palindrome: true
PS D:\study\SQA Course (Ostad)\5 Module\Module-5_Assignment_Programming_Code>
```

## Explanation:

- The function takes a word as input.
- It reverses the word using `split()`, `reverse()`, and `join()`.
- The reversed word is compared with the original word.
- If both are the same, the function returns `true`.
- Otherwise, it returns `false`.
- For example, "racecar" is a palindrome because it reads the same backward and forward.

## 4. Sum of Array Elements

Write a function that takes an array of numbers and returns the sum of all elements in the array.

### Source Code:

```
function sumArray(numbers) {  
    let sum = 0;  
    for (let i = 0; i < numbers.length; i++) {  
        sum = sum + numbers[i];  
    }  
    return sum;  
}  
  
let numbers = [10, 20, 30, 40];  
console.log("Array:", numbers);  
console.log("Sum:", sumArray(numbers));
```

### Input:

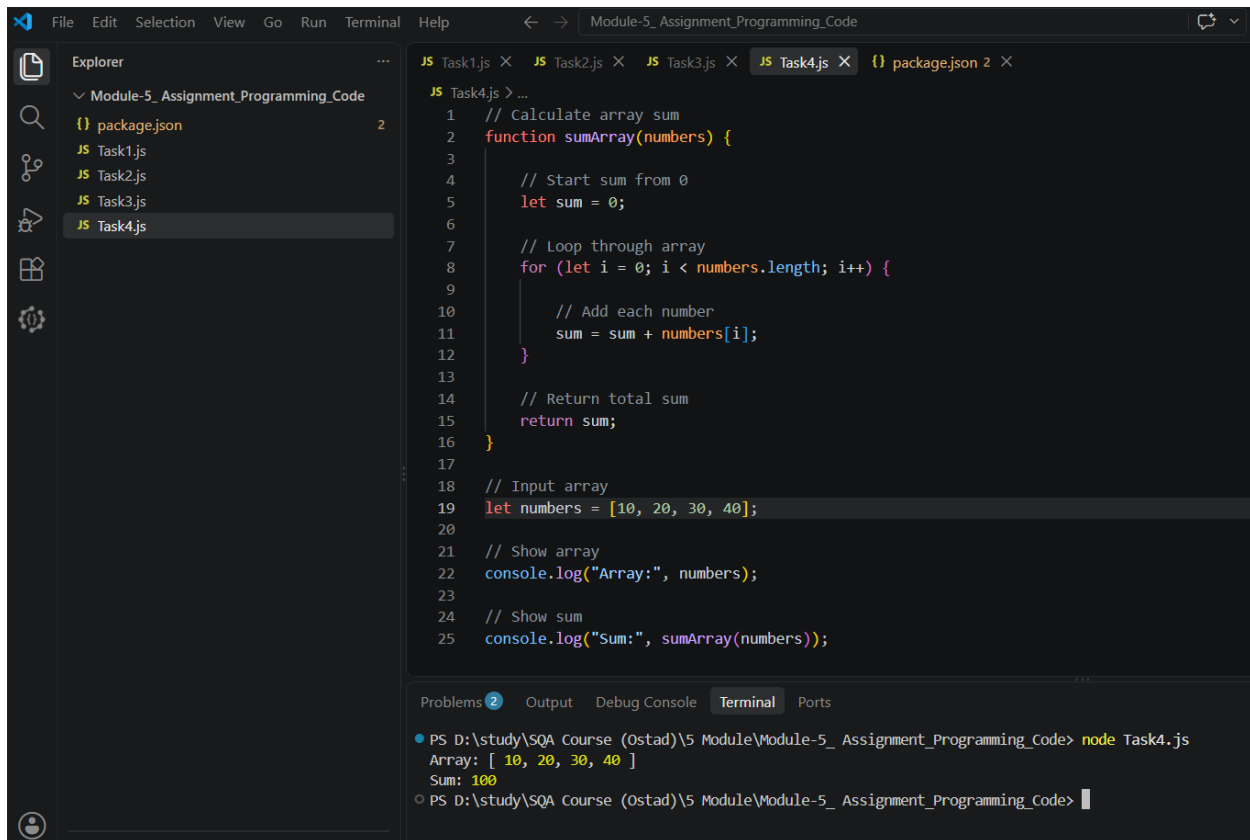
Array = [10, 20, 30, 40]

### Output:

Array: [ 10, 20, 30, 40 ]

Sum: 100

## Screenshot:



```
File Edit Selection View Go Run Terminal Help
Module-5_Assignment_Programming_Code

Explorer
Module-5_Assignment_Programming_Code
package.json
Task1.js
Task2.js
Task3.js
Task4.js

JS Task4.js > ...
1 // Calculate array sum
2 function sumArray(numbers) {
3
4     // Start sum from 0
5     let sum = 0;
6
7     // Loop through array
8     for (let i = 0; i < numbers.length; i++) {
9
10        // Add each number
11        sum = sum + numbers[i];
12    }
13
14    // Return total sum
15    return sum;
16 }
17
18 // Input array
19 let numbers = [10, 20, 30, 40];
20
21 // Show array
22 console.log("Array:", numbers);
23
24 // Show sum
25 console.log("Sum:", sumArray(numbers));

Problems 2 Output Debug Console Terminal Ports
PS D:\study\SQLA Course (Ostad)\5 Module\Module-5_Assignment_Programming_Code> node Task4.js
Array: [ 10, 20, 30, 40 ]
Sum: 100
PS D:\study\SQLA Course (Ostad)\5 Module\Module-5_Assignment_Programming_Code>
```

## Explanation:

- The function takes an array of numbers as input.
- It starts the sum variable with **0**.
- A for loop goes through each element of the array.
- Each number is added to the sum variable.
- Finally, the function returns the total sum.
- For example, **10 + 20 + 30 + 40 = 100**.

## 5. FizzBuzz (Basic)

Write a loop from 1 to 15 that prints "Fizz" for numbers divisible by 3, "Buzz" for numbers divisible by 5, "FizzBuzz" for numbers divisible by both, and the number itself otherwise.

### Source Code:

```
function fizzBuzz() {  
  for (let i = 1; i <= 15; i++) {  
    if (i % 3 === 0 && i % 5 === 0) {  
      console.log("FizzBuzz");  
    }  
    else if (i % 3 === 0) {  
      console.log("Fizz");  
    }  
    else if (i % 5 === 0) {  
      console.log("Buzz");  
    }  
    else {  
      console.log(i);  
    }  
  }  
}
```

fizzBuzz();

**Input:**

Numbers from 1 to 15

**Output:**

1

2

Fizz

4

Buzz

Fizz

7

8

Fizz

Buzz

11

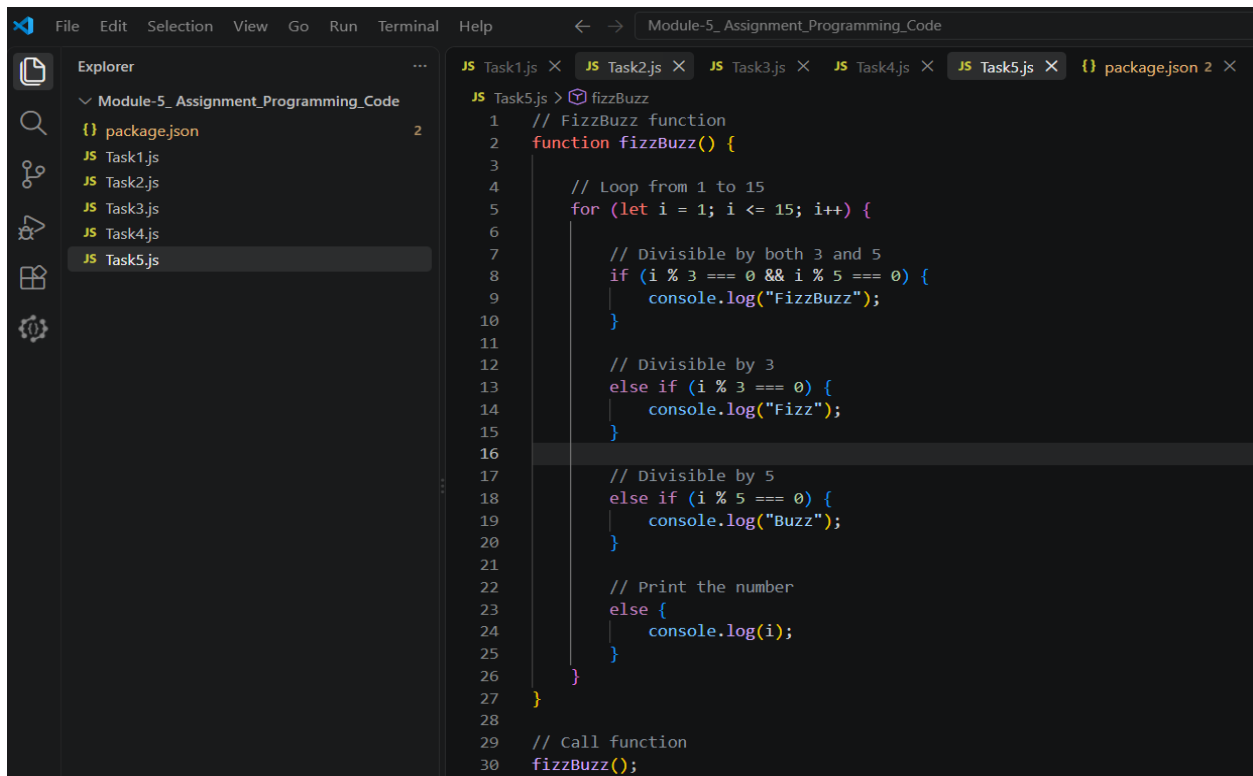
Fizz

13

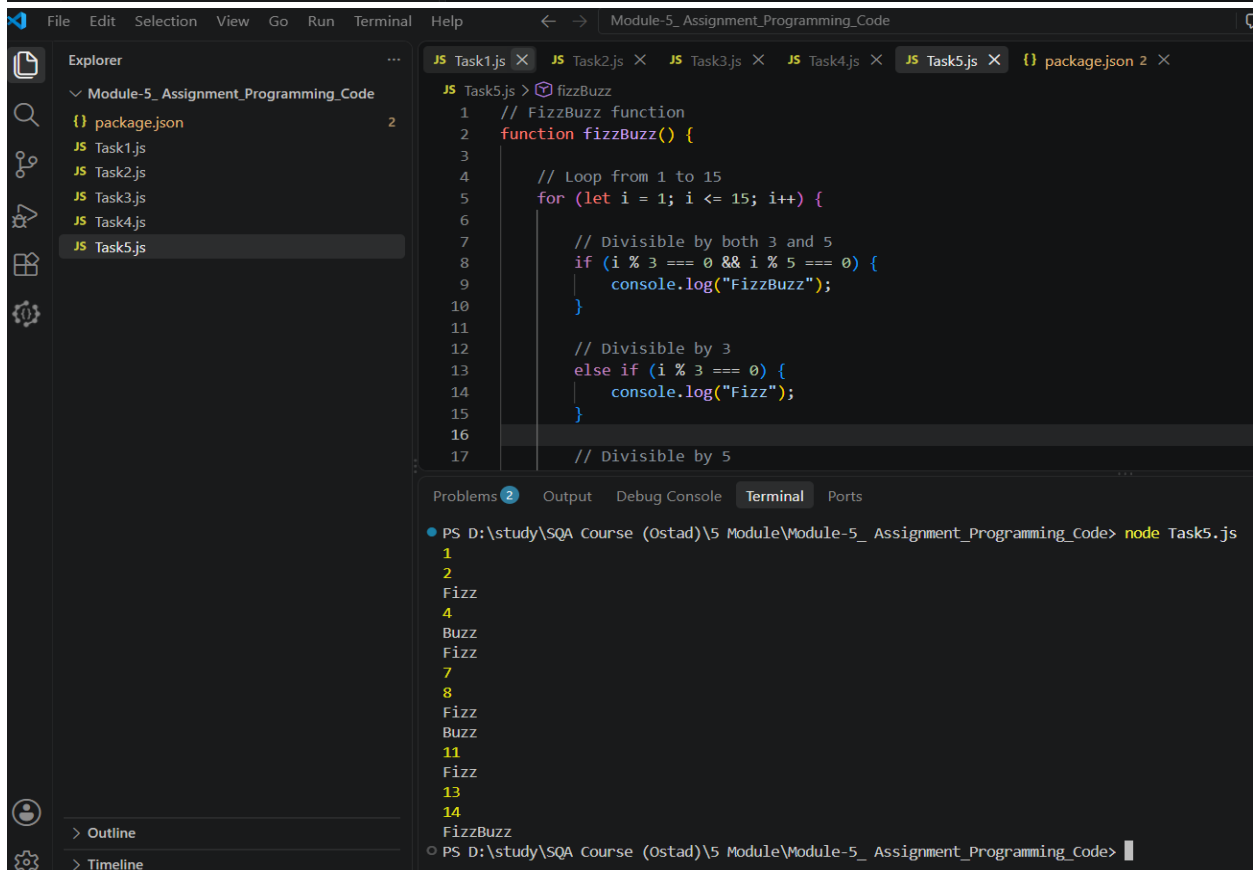
14

FizzBuzz

# Screenshot:



```
JS Task5.js > fizzBuzz
1 // FizzBuzz function
2 function fizzBuzz() {
3
4     // Loop from 1 to 15
5     for (let i = 1; i <= 15; i++) {
6
7         // Divisible by both 3 and 5
8         if (i % 3 === 0 && i % 5 === 0) {
9             console.log("FizzBuzz");
10        }
11
12        // Divisible by 3
13        else if (i % 3 === 0) {
14            console.log("Fizz");
15        }
16
17        // Divisible by 5
18        else if (i % 5 === 0) {
19            console.log("Buzz");
20        }
21
22        // Print the number
23        else {
24            console.log(i);
25        }
26    }
27 }
28
29 // Call function
30 fizzBuzz();
```



```
JS Task5.js > fizzBuzz
1 // FizzBuzz function
2 function fizzBuzz() {
3
4     // Loop from 1 to 15
5     for (let i = 1; i <= 15; i++) {
6
7         // Divisible by both 3 and 5
8         if (i % 3 === 0 && i % 5 === 0) {
9             console.log("FizzBuzz");
10        }
11
12        // Divisible by 3
13        else if (i % 3 === 0) {
14            console.log("Fizz");
15        }
16
17        // Divisible by 5
```

Problems 2 Output Debug Console Terminal Ports

```
● PS D:\study\SQA Course (Ostad)\5 Module\Module-5_ Assignment_Programming_Code> node Task5.js
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
○ PS D:\study\SQA Course (Ostad)\5 Module\Module-5_ Assignment_Programming_Code> |
```

## Explanation:

- The function uses a for loop to print numbers from **1 to 15**.
- The % operator is used to check whether a number is divisible by 3 or 5.
- If a number is divisible by both 3 and 5, it prints "**FizzBuzz**".
- If it is divisible by 3, it prints "**Fizz**".
- If it is divisible by 5, it prints "**Buzz**".
- For other numbers, it prints the number itself.